AutoCorp Hub – AI-Powered Multi-Agent Platform for Enterprise Automation

1.Team Members:

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2. Project Goals:

AutoCorp Hub's goal is to build a cloud-based AI platform where businesses can hire digital workers (AI agents) to run their daily business tasks.

A shared orchestrator connects all of the agents (like HR Onboarding, Mail Automation, Meeting Scheduler, and Jira Manager) into a single microservice.

Key objectives:

- Automate HR onboarding workflows (email parsing, calendar scheduling, and document generation).

- Build and deploy scalable microservices using **Google Kubernetes Engine (GKE)**.

- Use **Gmail API** for parsing onboarding or interview requests and triggering downstream workflows.

- Use **Google Calendar API** to automatically schedule and manage HR events and meetings.

- Develop a **Streamlit-based dashboard** for HR managers to monitor automation and control workflows.

- Integrate **LLM models** (through Hugging Face or Vertex AI) to interpret natural-language email content and extract structured onboarding data.

- Deploy, monitor, and scale the platform entirely within **Google Cloud Platform**.

3. Software and Hardware Components:

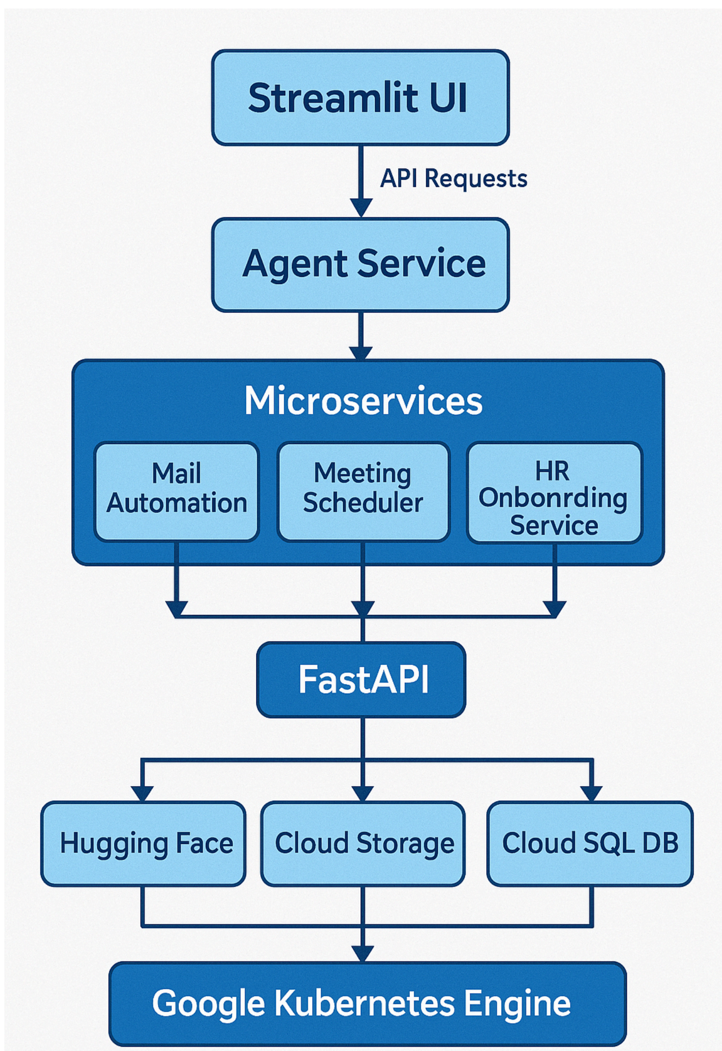
Software Components:

Component	Description
Frontend: Streamlit	Interface for subscribing to the micro agents and use them.
Backend Framework: FastAPI	REST microservices managing onboarding workflows, Gmail/Calender integration.
LLM Engine: Hugging Face API/Vertex AI	Process EMAIL content
Containerization: Docker	Packages each microservice for deployment on GKE.
Orchestration: GKE	Manages microservices , scaling, and networking.
Authentication: Google OAuth2	Used for secure access to Gmail and calender.
Storage: Google Cloud Storage(GCS)	Stores onboarding forms, resumes, and generated offer letters.

Hardware Components:

Resource	Description
Developer Machines	Local Laptops for development and testing
GCP Compute Nodes	Virtual machines for hosting GKE clusters.
GPU Node	For faster LLM inference and text extraction.

4. Architectural Diagram:



5. Interaction Between Components:

- **User Interaction(Streamlit UI):** The HR manager safely logs into the system with Google OAuth2.

  The user can choose agent services like Mail automation, Meeting scheduler, and HR Onboarding Service from the dashboard.

  The Agent service sends an API request to the backend when the user starts an automation.

- **Agent Service:** The streamlit UI sends API requests to the agent service. It figures out which microservice should do the job based on what the user chose.

- **Microservices Layer(Mail Automation, Meeting Scheduler, HR Onboarding):** uses the Gmail API to read emails and get the subject, body, attachments, and sender information. LLM for getting structured information out of data.

  Use the Calender API to automatically set up meetings or interviews based on information taken from emails. Sends calendar invites to people who are going to the event and changes the schedules on the fly.

  Makes onboarding documents like offer letters, welcome emails, and HR forms. Uploads files that were made to GCS.

- **Backend Orchestration (FastAPI):** FastAPI acts as main orchestration and routing engine for all microservices.

- **Infrastructure Orchestration(Google Kubernetes Engine):** All components are deployed as Docker containers within GKE.

6. Debugging, Training, and Testing Mechanisms:

- **Debugging Approach:**
  Use FastAPI to independently test every microservice. To track errors, use Python logging. To track inter-service latency and locate bottlenecks, turn on Google Cloud Trace.

- **Testing & Validation:**
  Pytest for API endpoints, LLM extraction functions, and DB interaction.
  Integration Tests: Validate end-to-end flow from Gmail->LLM->Calender->HR document generation.
  Load Testing: Use Locust to simulate concurrent HR users performing automation.
  LLM Validation: use a dataset of HR emails to evaluate extraction accuracy and model reliability.

7. Meeting Project Requirements:

AutoCorp Hub integrates four distinct Google Cloud technologies:

Cloud Service Category	GCP Technology Used	Purpose
Compute	Google Kubernetes Engine(GKE)	Hosts and Scales FastAPI microservices.
Storage	Google Cloud Storage(GCS)	Stores onboarding forms, resumes, and documents.
Database	Cloud SQL (PostgreSQL)	Maintains structured onboarding an user data.
API service	Gmail API + Google Calender API + Vertex AI	Automates HR Email handling, scheduling, and intelligent text processing.

Ambitions and Future Scope:

1. Project Ambitions:

   The main goal is to convert internal business processes and traditional HR operations into fully automated, intelligent, cloud-native workflows that reduce manual labor and increase productivity.

   Our goal with this project is to use scalable microservice-based architecture and AI-driven decision-making to create a modular platform that can manage various enterprise functions.

2. Future Scope:

   Extending the use of Micro agents to additional divisions, such as finance operations, Jira and task automation, payroll processing, and attendance and leave management.

   Expand deployment to AWS and AZURE to provide redundancy and scalability for hybrid clouds.

   Utilizing actual HR datasets to refine domain-specific LLMs in order to enhance intent classification and contextual understanding.

   Put in place a RAG system so that agents can consult legal documents, HR policies, and company policies.